

Module 5: Matrix Algebra

Ph.D. Math Camp

Linear Models and Matrix Algebra

Linear Models

- ▶ Thus far when solving for unknowns (e.g., P^* , Q^*) we have stuck with two equations and two unknowns.
- ▶ Including more unknowns and equations makes solutions cumbersome. In order to handle large systems of simultaneous **linear** equations we need *matrix algebra*

System of Equations

- Recall we had a system of two linear equations when we analyzed supply and demand

$$c_1q + c_2p = b$$

$$\gamma_1q - \gamma_2p = d$$

- In general, a system of n linear equations in n variables $(x_1, x_2, x_3, \dots, x_n)$ can be arranged in the following format

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = d_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = d_2$$

.....

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = d_m$$

Matrices as Arrays

There are three components in the system of equations (above).

- ▶ The set of coefficients a_{ij} .
- ▶ The set of variables $x_1, x_2, x_3, \dots, x_n$
- ▶ The constant terms $d_1, d_2, \dots, d_m$

Matrices as Arrays

- ▶ As an example, given the linear system of equations

$$6x_1 + 3x_2 + x_3 = 22$$

$$x_1 + 4x_2 - 2x_3 = 12$$

$$4x_1 - x_2 + 5x_3 = 10$$

we can write

$$A = \begin{bmatrix} 6 & 3 & 1 \\ 1 & 4 & -2 \\ 4 & -1 & 5 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad d = \begin{bmatrix} 22 \\ 12 \\ 10 \end{bmatrix}$$

- ▶ Each of these three arrays constitutes a matrix.

Matrices as Arrays

- ▶ **Matrix:** A rectangular array of numbers, parameters, or variables. The numbers of the array, referred to as *elements* of the matrix, are usually enclosed in brackets.
- ▶ Note that the elements are separated by spaces, not commas. As a shorthand, the array in matrix A can be written more simply as

$$A = [a_{ij}] \quad (i = 1, 2, \dots, m ; j = 1, 2, \dots, n)$$

Vectors as Special Matrices

- ▶ The number of rows and the number of columns in a matrix together define the *dimension* of the matrix. Since matrix A contains m rows and n columns, it is said to be of dimension $m \times n$ (read “m by n”). To be consistent, the row number should always precede the column number.
- ▶ In the special case where $m = n$, the matrix is called a *square matrix*. Thus, matrix A is a 3×3 square matrix.
- ▶ Some matrices may contain only one column, such as x and d . These matrices are called *column vectors*.
- ▶ The dimension of x is $n \times 1$ and d is $m \times 1$.

- ▶ If we arranged the variables x_j in a horizontal array, the result should be a $1 \times n$ matrix, which is called a *row vector*. For notation purposes, a row vector is often distinguished from a column vector by the use of a primed symbol:

$$x' = [x_1 \ x_2 \ x_3 \ \dots \ x_n]$$

- ▶ With matrices defined, we can express an equation system as follows

$$Ax = d$$

- ▶ No matter how extensive the system is, we can express it this way.

Matrix Operations

- Note that **equality** means that two matrices are equal iff they have the same dimensions and have identical elements.

$$\begin{bmatrix} 4 & 3 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 2 & 0 \end{bmatrix} \neq \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}$$

- As another example, if $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$ means that $x = 7$ and $y = 4$.

Addition and Subtraction

- ▶ Two matrices can be added iff they have the same dimension. When this dimensional requirement is met, the matrices are said to be *conformable for addition*.
- ▶ In that case, the addition of $A = [a_{ij}]$ and $B = [b_{ij}]$ is defined as the addition of each pair of corresponding elements.
- ▶ Generally, we can state the rule as $[a_{ij}] + [b_{ij}] = [c_{ij}]$ where $c_{ij} = a_{ij} + b_{ij}$. Note that $[c_{ij}]$ will have the same dimension as the matrices $[a_{ij}]$ and $[b_{ij}]$.

Addition and Subtraction

- ▶ The **subtraction** operation $A - B$ can similarly be defined if and only if A and B have the same dimension. The operation entails the results

$$[a_{ij}] - [b_{ij}] = [d_{ij}] \quad \text{where} \quad d_{ij} = a_{ij} - b_{ij}$$

Scalar Multiplication

- ▶ To multiply a matrix by a number– or in matrix-algebra terminology, by a *scalar*– is to multiply **every** element of that matrix by the given scalar. Thus, one “scales up (or down)”

Matrix Algebra

1. $A + B$, where $A = \begin{bmatrix} 4 & 9 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$

2. $A - B$, where $\begin{bmatrix} 19 & 3 \\ 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 6 & 8 \\ 1 & 3 \end{bmatrix}$

3. $7 \begin{bmatrix} 3 & -1 \\ 0 & 5 \end{bmatrix}$

4. $\frac{1}{2} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

5. $A + (-1)B$, where $A = \begin{bmatrix} 4 & 9 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$

Matrix Operations (Continued)

Multiplication of Matrices

- ▶ The multiplication of two matrices is contingent upon the satisfaction of a different dimensional requirement.
- ▶ Suppose that, given two matrices A and B , we want to find the product AB .
- ▶ The *conformability condition* for multiplication is that the *column* dimension of A (the “lead” matrix in AB) must equal the row dimension of B (the “lag” matrix).

Multiplication of Matrices

- ▶ For instance,

$$\underset{(1 \times 2)}{A} = [a_{11} \ a_{12}] \quad \underset{(2 \times 3)}{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

- ▶ The product AB then is defined, since A has *two columns* and B has *two rows*.
- ▶ You can check this quickly by checking the second number in the dimension indicator for A and the first number in the dimension indicator for B .
- ▶ Conversely, BA is not defined.

Multiplication of Matrices

$$\underset{(1 \times 2)}{A} = [a_{11} \ a_{12}] \quad \underset{(2 \times 3)}{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

- ▶ Generally, if A is of dimension $m \times n$ and B is of dimension $p \times q$, the matrix product will be defined iff $n = p$.
- ▶ If defined, the product of AB will have dimension $m \times q$. In this example, AB will have dimension 1×3 .
- ▶ Thus, the resulting matrix (call it C) will take the form

$$AB = C = [c_{11} \ c_{12} \ c_{13}]$$

- ▶ c_{ij} is defined as a sum of the products, to be computed from the elements in the i th row of A , and those in the j th column of B .

Multiplication of Matrices

$$AB = C = [c_{11} \ c_{12} \ c_{13}]$$

- For instance, to find c_{11} we take the first row in A and the first column in B and pair the elements sequentially, multiply out each pair, and take the sum of the resulting products, to get

$$c_{11} = a_{11}b_{11} + a_{12}b_{21}$$

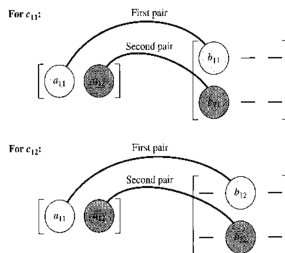
Similarly, for c_{12}

$$c_{12} = a_{11}b_{12} + a_{12}b_{22}$$

And for

$$c_{13} = a_{11}b_{13} + a_{12}b_{23}$$

Multiplication of Matrices



- ▶ The multiplication procedure illustrated in the figure can also be described using the concept of *inner product* of two vectors.
- ▶ Given two vectors u and v with n elements each, say, $(u_1, u_2, u_3, \dots, u_n)$ and $(v_1, v_2, \dots, v_n)$, arranged as *either* two rows or as two columns or as one row and one column, their inner product written as $u \cdot v$ is defined as

$$u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

Multiplication of Matrices

Try some examples. Find AB

1. $A = \begin{bmatrix} 1 & 3 \\ 2 & 8 \\ 4 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$

2. $A = \begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & 3 \\ 4 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -\frac{1}{5} & \frac{3}{10} \\ -1 & \frac{1}{5} & \frac{7}{10} \\ 0 & \frac{2}{5} & -\frac{1}{10} \end{bmatrix}$

Multiplication of Matrices

1.

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 8 \\ 4 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

$$\begin{aligned} AB &= \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} \\ a_{21}b_{11} + a_{22}b_{21} \\ a_{31}b_{11} + a_{32}b_{21} \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 3 \cdot 9 \\ 2 \cdot 5 + 8 \cdot 9 \\ 4 \cdot 5 + 0 \cdot 9 \end{bmatrix} \\ &= \begin{bmatrix} 5 + 27 \\ 10 + 72 \\ 20 + 0 \end{bmatrix} = \begin{bmatrix} 32 \\ 82 \\ 20 \end{bmatrix} \end{aligned}$$

Matrix Multiplication

2.

$$A = \begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & 3 \\ 4 & 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -\frac{1}{5} & \frac{3}{10} \\ -1 & \frac{1}{5} & \frac{7}{10} \\ 0 & \frac{2}{5} & -\frac{1}{10} \end{bmatrix}$$

$$C = \begin{bmatrix} 3 \cdot 0 + (-1) \cdot (-1) + 2 \cdot 0 & 3 \cdot \left(-\frac{1}{5}\right) + (-1) \cdot \frac{1}{5} + 2 \cdot \frac{2}{5} & 3 \cdot \frac{3}{10} + (-1) \cdot \frac{7}{10} + 2 \cdot \left(-\frac{1}{10}\right) \\ 1 \cdot 0 + 0 \cdot (-1) + 3 \cdot 0 & 1 \cdot \left(-\frac{1}{5}\right) + 0 \cdot \frac{1}{5} + 3 \cdot \frac{2}{5} & 1 \cdot \frac{3}{10} + 0 \cdot \frac{7}{10} + 3 \cdot \left(-\frac{1}{10}\right) \\ 4 \cdot 0 + 0 \cdot (-1) + 2 \cdot 0 & 4 \cdot \left(-\frac{1}{5}\right) + 0 \cdot \frac{1}{5} + 2 \cdot \frac{2}{5} & 4 \cdot \frac{3}{10} + 0 \cdot \frac{7}{10} + 2 \cdot \left(-\frac{1}{10}\right) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$AB = I_3$$

Matrix Multiplication

$$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$AB = I_3$$

- ▶ This square matrix with 1s in its *principal diagonal* and 0s everywhere else exemplifies an important type of matrix known as the *identity matrix*.

- ▶ Let's revisit the following system of equations. Write out as a product of matrices.

$$6x_1 + 3x_2 + x_3 = 22$$

$$x_1 + 4x_2 - 2x_3 = 12$$

$$4x_1 - x_2 + 5x_3 = 10$$



$$Ax = \begin{bmatrix} 6 & 3 & 1 \\ 1 & 4 & -2 \\ 4 & -1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 22 \\ 12 \\ 10 \end{bmatrix} = d$$

$(3 \times 3) \qquad (3 \times 1) \qquad (3 \times 1)$

- ▶ Note: To use the matrix notation $Ax = d$, it is necessary, because of the conformability condition, to arrange the x_j s into a column vector, even though they are listed in horizontal order.

- ▶ $AB = C = [c_{11} \ c_{12} \ c_{13}]$
- ▶ We can also use the summation shorthand to write out the matrix multiplication discussed above.

$$c_{11} = a_{11}b_{11} + a_{12}b_{21} = \sum_{k=1}^2 a_{1k}b_{k1} \quad (1)$$

$$c_{12} = a_{11}b_{12} + a_{12}b_{22} = \sum_{k=1}^2 a_{1k}b_{k2} \quad (2)$$

$$c_{13} = a_{11}b_{13} + a_{12}b_{23} = \sum_{k=1}^2 a_{1k}b_{k3} \quad (3)$$

- ▶ In each case, the first subscript of c_{1j} is reflected in the first subscript of a_{1k} , and the second subscript of c_{1j} is reflected in the second subscript of b_{kj} in the $\sum$ expression.
- ▶ The index k is a dummy subscript and serves to indicate which particular pair of elements is being multiplied.

- Extending this to the multiplication of an $m \times n$ matrix $A = [a_{ik}]$ and an $n \times p$ matrix $B = [b_{kj}]$, we may now write the elements of the $m \times p$ product matrix $AB = C = [c_{ij}]$ as

$$c_{11} = \sum_{k=1}^n a_{1k} b_{k1} \quad c_{12} = \sum_{k=1}^n a_{1k} b_{k2} \dots$$

or more generally,

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} \quad \begin{pmatrix} i = 1, 2, \dots, m \\ j = 1, 2, \dots, p \end{pmatrix}$$

Additional Vector Operations

An $m \times 1$ column vector u , and a $1 \times n$ row vector v' , yield a product matrix uv' of dimension $m \times n$. Suppose $u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $v' = [1 \ 4 \ 5]$

$$uv' = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \times [1 \ 4 \ 5] = \begin{bmatrix} 3 \times 1 & 3 \times 4 & 3 \times 5 \\ 2 \times 1 & 2 \times 4 & 2 \times 5 \end{bmatrix} = \begin{bmatrix} 3 & 12 & 15 \\ 2 & 8 & 10 \end{bmatrix}$$

Additional Vector Operations

- ▶ On the other hand if we were interested in multiplying

$$u' = [3 \quad 4] \text{ and } v = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$$

$$u'v = [3(9) + 4(7)] = [55]$$

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- ▶ As another example suppose $u' = [3 \ 6 \ 9]$. Find $u'u$.

$$u'u = [3 \ 6 \ 9] \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = [(3)^2 + (6)^2 + (9)^2] = [126]$$

Additional Vector Operations

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$$u'u = [3 \quad 6 \quad 9] \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = [(3)^2 + (6)^2 + (9)^2] = [126]$$

- ▶ Note that $u'u$ is not the same as uu' . The latter will result in a matrix larger than 1×1 .

Geometric Interpretations of Vector Operations

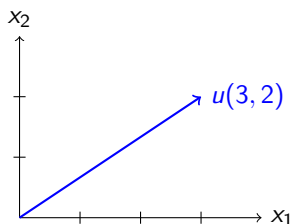
- ▶ A column or row vector can also be viewed as a point in space. Suppose we plot the point $(3,2)$ in a 2-space and label it u .

Geometric Interpretations of Vector Operations

- ▶ A column or row vector can also be viewed as a point in space. Suppose we plot the point $(3,2)$ in a 2-space and label it u .
- ▶ This is the geometric counterpart of the vector $u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ or the vector $u' = [3 \ 2]$. Both indicate the same ordered pair.

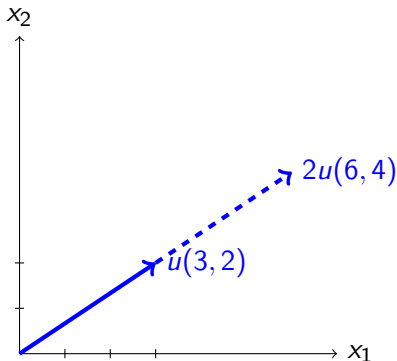
Geometric Interpretations of Vector Operations

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- ▶ This is the geometric counterpart of the vector $u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ or the vector $u' = [3 \ 2]$. Both indicate the same ordered pair.
- ▶ If an arrow is drawn from the point of the origin $(0,0)$ to the point u , it will specify the unique straight route by which to reach the destination point u from the point of origin. Such an arrow, with a definite length and direction, is called a *radius vector*.



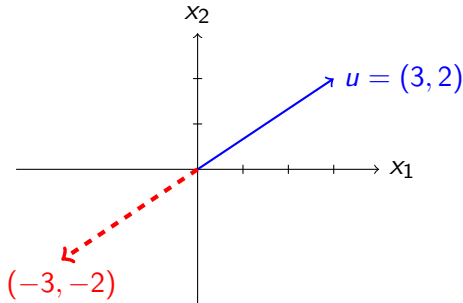
Geometric Interpretations of Vector Operations

- ▶ We can consider scalar multiplication, addition and subtraction of vectors, and linear combinations of vectors.
- ▶ For instance, if we plot $2u = 2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$



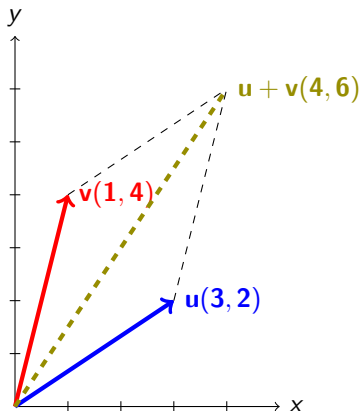
Geometric Interpretations of Vector Operations

- ▶ Multiplication of a vector u by any scalar k will produce an overlapping arrow where the arrowhead will be relocated, unless $k = 1$. If $k = 0$, the arrow will shrink into the point of the origin representing a null vector $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- ▶ A negative scalar multiplier will reverse the direction of the arrow.



Geometric Interpretations of Vector Operations

- ▶ We can also add vectors. Consider $v = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ and $u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. The sum is $u + v = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$
- ▶ We can do the same with vector differences.



Linear Dependence

- ▶ A set of vectors $v_1, \dots, v_n$ is said to be *linearly dependent* if (and only if) any one of them can be expressed as a linear combination of the remaining vectors; otherwise, they are *linearly independent*.

Linear Dependence

- ▶ A set of vectors $v_1, \dots, v_n$ is said to be *linearly dependent* if (and only if) any one of them can be expressed as a linear combination of the remaining vectors; otherwise, they are *linearly independent*.
- ▶ For example, take the vectors $v_1 = \begin{bmatrix} 2 \\ 7 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$. These are linearly dependent because v_3 is a linear combination of v_1 and v_2 :

$$3v_1 - 2v_2 = \begin{bmatrix} 6 \\ 21 \end{bmatrix} - \begin{bmatrix} 2 \\ 16 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} = v_3$$

We can also express this equation as $3v_1 - 2v_2 - v_3 = 0$.

Where $0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is a zero vector.

Linear Dependence

- ▶ Another example of linearly dependent vectors: $v'_1 = [5 \ 12]$ and $v'_2 = [10 \ 24]$.

$$2v'_1 = 2[5 \ 12] = [10 \ 24] = v'_2$$

As before, this equation can be written as $2v'_1 - v'_2 = 0'$. $0'$ is a null row vector

Linear Dependence

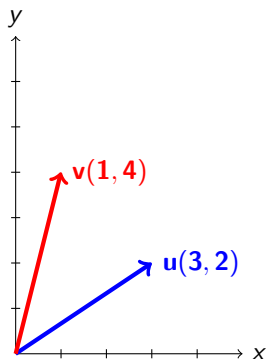
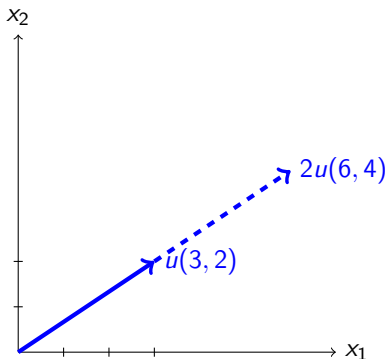
- ▶ We can thus redefine linear dependence as follows: A set of m -vectors $v_1, \dots, v_n$ is linearly dependent if and only if there exists a set of scalars $k_1, \dots, k_n$ (not all zero) such that

$$\sum_{i=1}^n k_i v_i = \underset{m \times 1}{0}$$

- ▶ If $k_i = 0$ is the only set of values that can satisfy this equality then we say that these vectors are *linearly independent*.

Linear Dependence

- We can also see this geometrically. u and $2u$ are dependent, while u and v are independent since we cannot express one as a multiple of the other. Their arrows do not lie on a single straight line.



Vector Space

- ▶ It turns out that by extending, shortening, and reversing a given independent vectors u and v and combining them, we can generate an infinite number of new vectors. We call this the two-dimensional *vector space*.

Vector Space

- ▶ It turns out that by extending, shortening, and reversing a given independent vectors u and v and combining them, we can generate an infinite number of new vectors. We call this the two-dimensional *vector space*.
- ▶ Two linearly independent vectors u and v can be a basis for 2-space. Consider two vectors $[1 \ 0]$ and $[0 \ 1]$, which are called unit vectors. The first one plots an arrow lying along the horizontal axis, and the second an arrow lying along the vertical axis. Since they are linearly independent they can serve as the basis for 2-space.

Vector Space

- ▶ By analogy, the three-dimensional vector space is the totality of 3-vectors, and it must be spanned by exactly three linearly independent 3-vectors. Consider the set of three unit vectors:

$$e_1 \equiv \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad e_2 \equiv \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad e_3 \equiv \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

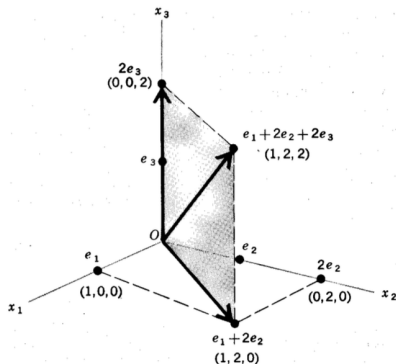
- ▶ These arrows lie on the three axes of the 3-space. We can generate the entire 3-space from these vectors. For instance,

take the vector $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

- ▶ This vector can be considered the linear combination $e_1 + 2e_2 + 2e_3$.

Vector Space

- This vector can be considered the linear combination $e_1 + 2e_2 + 2e_3$.



Vector Space

- ▶ We can extend this to n -space (although not graphable). This n -space is sometimes referred to as *Euclidean n -space*.
- ▶ For any pair of vectors u and v in a given space, the distance from u to v is some real-valued function

$$d(u, v) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2}$$

- ▶ For example, the distance between points $(6,4)$ and $(3,2)$ can be calculated as follows

$$\sqrt{(6 - 3)^2 + (4 - 2)^2} = \sqrt{3^2 + 2^2} = \sqrt{13}$$

Laws of Operations

- ▶ Commutative Law: $A + B = B + A$
- ▶ Associative Law: $(A + B) + C = A + (B + C)$

Given $v_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $v_2 = \begin{bmatrix} 9 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$, we find that

$$(v_1 + v_2) - v_3 = \begin{bmatrix} 12 \\ 5 \end{bmatrix} - \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

Which is equal to

$$v_1 + (v_2 - v_3) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 7 \\ -4 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

Laws of Operations

- ▶ Matrix multiplication is **not** commutative

$$AB \neq BA$$

- ▶ If AB is defined, that does not mean BA is defined. Even if both are, the general rule applies $AB \neq BA$.

Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 6 & 7 \end{bmatrix}$

$$AB = \begin{bmatrix} 1(0) + 2(6) & 1(-1) + 2(7) \\ 3(0) + 4(6) & 3(-1) + 4(7) \end{bmatrix} = \begin{bmatrix} 12 & 13 \\ 24 & 25 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0(1) + (-1)(3) & 0(2) + (-1)(4) \\ 6(1) + 7(3) & 6(2) + 7(4) \end{bmatrix} = \begin{bmatrix} -3 & -4 \\ 27 & 40 \end{bmatrix}$$

Laws of Operations

- ▶ Note that scalar multiplication of a matrix does obey the commutative law

$$kA = Ak$$

- ▶ Associative Law: $(AB)C = A(BC) = ABC$.
 - In forming the product ABC , the conformability condition must naturally be satisfied by each *adjacent* pair of matrices. If A is $m \times n$ and C is $p \times q$, then conformability requires B to be $n \times p$.

$$\begin{matrix} A & B & C \\ (m \times n) & (n \times p) & (p \times q) \end{matrix}$$

Laws of Operations

Find $C = AB$

1. $A = \begin{bmatrix} 12 & 14 \\ 20 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 9 \\ 0 & 52 \end{bmatrix}$

2. $A = \begin{bmatrix} 4 & 7 \\ 9 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 8 & 5 \\ 2 & 6 & 7 \end{bmatrix}$

3. $A = \begin{bmatrix} 7 & 11 \\ 2 & 9 \\ 10 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 12 & 4 & 5 \\ 3 & 6 & 1 \end{bmatrix}$

4. $A = \begin{bmatrix} 4 & 7 \\ 9 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

5. $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 7 \\ 9 & 1 \end{bmatrix}$

Identity Matrices

- ▶ An identity matrix is a **square** matrix with 1s in its principal diagonals and 0s everywhere else. It is denoted by the symbol I or I_n . Where n indicates the dimension.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- ▶ The importance of this special type of matrix lies in the fact that it plays a role similar to that of the number 1 in scalar algebra. For any number a , we have $1(a) = a(1) = a$. Similarly, for any matrix A , we have

$$IA = AI = A$$

Identity Matrices

- ▶ As an example, suppose $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{bmatrix}$

$$IA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{bmatrix} = A$$

- ▶ Similarly,

$$AI = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{bmatrix} = A$$

- ▶ *Premultiplication* and *postmultiplication* of A by I would call for identity matrices of different dimensions to abide by conformability.

Identity Matrices

- ▶ The special nature of the identity matrices makes it possible, during the multiplication process, to *insert* or *delete* an identity matrix without affecting the matrix product. For instance,

$$AIB = \begin{matrix} A & I & B \\ (m \times n) & (n \times n) & (n \times p) \end{matrix} = (AI)B = \begin{matrix} A & B \\ (m \times n) & (n \times p) \end{matrix}$$

which shows that the presence or absence of I does not affect the product.

- ▶ An interesting case also occurs when $A = I_n$

$$AI_n = (I_n)^2 = I_n$$

- ▶ The identity matrix squared is equal to itself. Generally, an identity matrix remains unchanged when it is multiplied by itself any number of times.

Null Matrices

- ▶ Just as an identity matrix I plays the role of the number 1, a *null matrix*– or *zero matrix*– denoted by 0 , plays the role of the number 0. A null matrix is simply a matrix whose elements are all zero. This matrix is not restricted to being a square.
- ▶ Thus we can write

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

As an example,

$$A + 0 = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = A$$

Null Matrices

- ▶ However,

$$\begin{matrix} A & 0 \\ (2 \times 3) & (3 \times 1) \end{matrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- ▶ Note that the resulting null matrix is 2×1 and not 3×1

Null Matrices

- ▶ Null matrices observe the following rules regarding multiplication and addition

$$\begin{matrix} A \\ (m \times n) \end{matrix} + \begin{matrix} 0 \\ (m \times n) \end{matrix} = \begin{matrix} 0 \\ (m \times n) \end{matrix} + \begin{matrix} A \\ (m \times n) \end{matrix} = \begin{matrix} A \\ (m \times n) \end{matrix}$$

$$\begin{matrix} A \\ (m \times n) \end{matrix} \begin{matrix} 0 \\ (n \times p) \end{matrix} = \begin{matrix} 0 \\ (m \times p) \end{matrix} \quad \text{and} \quad \begin{matrix} 0 \\ (q \times m) \end{matrix} \begin{matrix} A \\ (m \times n) \end{matrix} = \begin{matrix} 0 \\ (q \times n) \end{matrix}$$

A Couple More Differences from Scalar Algebra

- ▶ We have seen that $AB \neq BA$
- ▶ In scalar algebra $ab = 0 \implies a = 0$ or $b = 0$. This is not so in matrix multiplication.

$$AB = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & 4 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

- ▶ For scalars, the equation $cd = ce$ ($c \neq 0$) $\implies d = e$. This does not always hold for matrices. Given

$$C = \begin{bmatrix} 2 & 3 \\ 6 & 9 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \quad E = \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix}$$

We find that

$$CD = CE = \begin{bmatrix} 5 & 8 \\ 15 & 24 \end{bmatrix}$$

even though $D \neq E$.

Transposes and Inverses

Transpose

- ▶ When the rows and columns of a matrix A are interchanged– so that its first row becomes the first column, and vice versa– we obtain the *transpose* of A . We denote this as A' or A^T .

- ▶ For example, given $A_{(2 \times 3)} = \begin{bmatrix} 3 & 8 & -9 \\ 1 & 0 & 4 \end{bmatrix}$ and $B_{(2 \times 2)} = \begin{bmatrix} 3 & 4 \\ 1 & 7 \end{bmatrix}$

We can interchange the rows and columns to get $A'_{(3 \times 2)} = \begin{bmatrix} 3 & 1 \\ 8 & 0 \\ -9 & 4 \end{bmatrix}$

and $B^T_{(2 \times 2)} = \begin{bmatrix} 3 & 1 \\ 4 & 7 \end{bmatrix}$

- ▶ By definition, a matrix A is $m \times n$; its transpose A' must be $n \times m$. However, an $n \times n$ square matrix possesses a transpose with the same dimension.

Transpose

Example: Find the transpose of each

1. $C = \begin{bmatrix} 9 & -1 \\ 2 & 0 \end{bmatrix}$

2. $D = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 3 & 7 \\ 4 & 7 & 2 \end{bmatrix}$

Transpose

- ▶ In each of these examples, the dimension of each transpose is identical to that of the original matrix. In D' , we also note the remarkable result that D' inherits not only the dimension of D but also the original array of elements.
- ▶ The fact that $D' = D$ is the result of the symmetry of the elements with reference to the principal diagonal. The principal diagonal in D is like a mirror.
- ▶ The matrix D exemplifies the special class of square matrices known as *symmetric matrices*. Another example of this is the identity matrix, $I' = I$.

Properties of Transposes

- ▶ $(A')' = A$
- ▶ $(A + B)' = A' + B'$. Transpose of the sum is the sum of the transpose.
- ▶ $(AB)' = B'A'$. The transpose of the product is the product of the transpose in reverse order.

Properties of Transposes

1. Show that $(A + B)' = A' + B'$ for $A = \begin{bmatrix} 4 & 1 \\ 9 & 0 \end{bmatrix}$ and

$$B = \begin{bmatrix} 2 & 0 \\ 7 & 1 \end{bmatrix}$$

2. Show that $(AB)' = B'A'$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and

$$B = \begin{bmatrix} 0 & -1 \\ 6 & 7 \end{bmatrix}$$

Inverses and Their Properties

- ▶ The *inverse* of a matrix may or may not exist.
- ▶ The inverse of a matrix A , denoted by A^{-1} , is defined only if A is a square matrix, in which case the inverse is the matrix that satisfies the condition:

$$AA^{-1} = A^{-1}A = I$$

Whether you pre-or-post multiply by A^{-1} , the product will be the identity matrix.

Inverses and Their Properties

- ▶ Not every square matrix has an inverse— squareness is *necessary* but not *sufficient*. If a square matrix has an inverse, it is said to be *nonsingular*. If A has no inverse, it is called *singular*.
- ▶ If A^{-1} exists, then the matrix A can be regarded as the inverse of A^{-1} . They are inverses of each other.

Inverses and Their Properties

If an inverse exists, it is unique. We can show this.

- ▶ Suppose B is the inverse of A . The claim is that $BA = AB = I$
 - Suppose there is another matrix C such that $CA = AC = I$
 - Take $AB = I$ and pre-multiply by $C \implies CAB = CI$
 - $\implies CAB = C$
 - Note that C is the inverse of $A \implies CA = I$
 - $\implies IB = C \Rightarrow B = C$
 - Thus, inverse is unique

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Inverses and Their Properties

$(AB)^{-1} = B^{-1}A^{-1}$. The inverse of the product is the product of the inverses in reverse.

► Suppose C is the inverse of $AB \Rightarrow C = (AB)^{-1}$ so that $C(AB) = I$

– Post-multiply by $B^{-1}A^{-1}$

– $\Rightarrow C(AB)B^{-1}A^{-1} = I \cdot B^{-1}A^{-1}$

$$\Rightarrow CA(BB^{-1})A^{-1} = B^{-1}A^{-1}$$

$$\Rightarrow CAA^{-1} = B^{-1}A^{-1}$$

$$\Rightarrow C = B^{-1}A^{-1}$$

► Thus, $C = B^{-1}A^{-1}$ and $C = (AB)^{-1}$

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$(A')^{-1} = (A^{-1})'$. The inverse of the transpose is the transpose of the inverse.

- ▶ Given that A' , let $D = (A')^{-1} \Rightarrow DA' = I$ (since D is the inverse of the transpose).
- ▶ Recall that we can express I as

$$(AA^{-1})' = I' = I$$

- ▶ Plug $I = (AA^{-1})'$ into $DA' = I \Rightarrow DA' = (AA^{-1})'$
- ▶ From before, recall that $(AB)' = B'A'$
 - $\Rightarrow DA' = (AA^{-1})' \Rightarrow DA' = (A^{-1})'A'$
- ▶ Post multiply by $(A')^{-1}$

$$\Rightarrow DA'(A')^{-1} = (A^{-1})'A'(A')^{-1}$$

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Solutions to Linear Systems of Equations

- ▶ We can now apply the concept of an inverse matrix to a simultaneous system of equations. Recall our example,

$$6x_1 + 3x_2 + x_3 = 22$$

$$x_1 + 4x_2 - 2x_3 = 12$$

$$4x_1 - x_2 + 5x_3 = 10$$

Solutions to Linear Systems of Equations

$$6x_1 + 3x_2 + x_3 = 22$$

$$x_1 + 4x_2 - 2x_3 = 12$$

$$4x_1 - x_2 + 5x_3 = 10$$

► We can write

$$A = \begin{bmatrix} 6 & 3 & 1 \\ 1 & 4 & -2 \\ 4 & -1 & 5 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad d = \begin{bmatrix} 22 \\ 12 \\ 10 \end{bmatrix}$$

Solutions to Linear Systems of Equations

- ▶ If the inverse of the matrix A exists, the premultiplication of both sides yields

$$A^{-1}Ax = A^{-1}d \implies \underset{(3 \times 1)}{x} = \underset{(3 \times 3)}{A^{-1}} \underset{(3 \times 1)}{d}$$

- ▶ The LHS is a column vector of variables and the RHS a product of known numbers. This allows us to find the solution for the x -values.
- ▶ Since A^{-1} is unique, $A^{-1}d$ must be a unique vector of

solutions. For this example $A^{-1} = \frac{1}{52} \begin{bmatrix} 18 & -16 & -10 \\ -13 & 26 & 13 \\ -17 & 18 & 21 \end{bmatrix}$

Solutions to Linear Systems of Equations

Thus,

$$\begin{bmatrix} x_1^* \\ x_2^* \\ x_3^* \end{bmatrix} = \frac{1}{52} \begin{bmatrix} 18 & -16 & -10 \\ -13 & 26 & 13 \\ -17 & 18 & 21 \end{bmatrix} \begin{bmatrix} 22 \\ 12 \\ 10 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

Plugging these values into the system should satisfy each equation.

Conditions for Non-Singularity

Non-Singularity

- ▶ Recall that a given coefficient matrix A can have an inverse (i.e., non-singular) only if it is square (necessary but not sufficient)
- ▶ A sufficient condition of non-singularity of a matrix is that the **rows** (columns) be **linearly independent**
- ▶ Together, (1) squareness; (2) linear independence $\implies$ Non-singularity.

Rows of a Matrix as Vectors

- Recall: an $n \times n$ coefficient matrix A can be viewed as an *ordered set of row vectors*.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1' \\ \mathbf{v}_2' \\ \vdots \\ \mathbf{v}_n' \end{bmatrix}$$

- Or a column vector whose elements themselves are row vectors

$$\mathbf{v}_i' = [a_{i1} \ a_{i2} \ \cdots \ a_{in}], \quad i = 1, 2, \dots, n.$$

Linear Independence of Rows

- ▶ For the rows to be linearly independent, *no row* can be written as a linear combination of the others.
- ▶ Linear row independence requires that the only set of scalars k_i which can satisfy

$$\sum_i^n k_i v'_i = 0$$

be $k_i = 0$ for $\forall i$

Linear Independence of Rows

For example

$$A = \begin{bmatrix} 3 & 4 & 5 \\ 0 & 1 & 2 \\ 6 & 8 & 10 \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1' \\ \mathbf{v}_2' \\ \mathbf{v}_3' \end{bmatrix}$$

- ▶ Since $\mathbf{v}_3' = k_1 \mathbf{v}_1' + k_2 \mathbf{v}_2'$ or

$$\begin{bmatrix} 6 & 8 & 10 \end{bmatrix} = 2 \begin{bmatrix} 3 & 4 & 5 \end{bmatrix} + 0 \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$$

- ▶ The third row can be expressed as a linear combination of the first two $\implies$ that rows are not linearly independent. $\mathbf{v}_3' = 2\mathbf{v}_1' + 0\mathbf{v}_2'$
- ▶ We can also express it as $2\mathbf{v}_1' + 0\mathbf{v}_2' - \mathbf{v}_3' = 0$
- ▶ As long as the set of scalars, k_i , that led to the zero vector are not all zero $\rightarrow$ rows are linearly dependent.

Why do we need linear independence?

- ▶ Usually, you hear that “you need as many equations as unknowns” to solve a system (e.g., two equations and two unknowns)– equations must be functionally independent
- ▶ This is why we work with a square matrix ($n \times n$)

- ▶ Take

$$\begin{bmatrix} 10 & 4 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

- ▶ Note that $v'_1 = 2v'_2 \implies$ dependence
- ▶ We don't know d_1 or d_2 , but there are two possibilities

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$$\begin{bmatrix} 10 & 4 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

- ▶ We don't know d_1 or d_2 , but there are two possibilities
 1. $d_1 = 2d_2$
 2. $d_1 \neq 2d_2$
- ▶ Consider (1), say $d_1 = 12, d_2 = 6$. The equations are consistent but linearly dependent. The first equation is two times the second.
- ▶ Thus, they are actually the same equation, with an infinite number of solutions that will satisfy both equations $\rightarrow$ no unique $x_1^*, x_2^* \rightarrow$ same line.

Why do we need linear independence?

$$\begin{bmatrix} 10 & 4 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

- ▶ We don't know d_1 or d_2 , but there are two possibilities
 1. $d_1 = 2d_2$
 2. $d_1 \neq 2d_2$
- ▶ Consider (2), say $d_1 = 12, d_2 = 0 \rightarrow$ equations are inconsistent
- ▶ If $10x_1 + 4x_2 = 12$ is true then $5x_1 + 2x_2 = 6$ must be true, but $d_2 = 0? \rightarrow$ inconsistency
- ▶ $\implies$ no unique solution.

Linear Independence

- ▶ For a square system, the only way to have a unique solution is to have linearly independent rows.
- ▶ If A is non-singular then A^{-1} exists
- ▶ If A^{-1} exists, then $x^* = A^{-1}d$ can be found

Rank of a Matrix

Rank of a matrix

- ▶ The number of linearly independent rows of a matrix is also called its rank
- ▶ To find the rank of a matrix we may have to transform it into what is called an *echelon matrix* by using one of three types of elementary row operations
 1. Interchange any two rows
 2. Multiply (or divide) a row by any scalar $k \neq 0$
 3. The addition of “ k times any row” to another row
- ▶ Although these operations change the matrix into a different form, none of them alter the rank.

Rank of a matrix

Let's find the rank of matrix $A = \begin{bmatrix} 0 & -11 & -4 \\ 2 & 6 & 2 \\ 4 & 1 & 0 \end{bmatrix}$

Rank of a matrix

1. Check the first column for zeros. If any row has a zero in the first column, move that row to the bottom: $R_1 \circlearrowleft R_3$

$$A_1 = \begin{bmatrix} 4 & 1 & 0 \\ 2 & 6 & 2 \\ 0 & -11 & -4 \end{bmatrix}$$

Rank of a matrix

2. Transform the first column of A_1 into a unit vector $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$\text{First, scale } R_1: R_1 = \frac{1}{4}R_1 \Rightarrow A_2 = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 2 & 6 & 2 \\ 0 & -11 & -4 \end{bmatrix}$$

Next, eliminate the 2 below the pivot (the 1 in R_1C_1):

$$R_2 = -2R_1 + R_2 \Rightarrow A_3 = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & \frac{11}{2} & 2 \\ 0 & -11 & -4 \end{bmatrix}$$

Rank of a matrix

3. Scale the $\frac{11}{2}$ in A_3 to make equal to 1

$$R_2 = \frac{2}{11}R_2 \Rightarrow A_4 = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & 1 & \frac{4}{11} \\ 0 & -11 & -4 \end{bmatrix}$$

4. Eliminate -11 in row 3

$$R_3 = 11R_2 + R_3 \Rightarrow A_5 = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & 1 & \frac{4}{11} \\ 0 & 0 & 0 \end{bmatrix}$$

Rank of a matrix

$$A_5 = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & 1 & \frac{4}{11} \\ 0 & 0 & 0 \end{bmatrix}$$

- ▶ We have now turned A into its *echelon* form or a triangular matrix.
 1. Non-zero rows appear above zero rows
 2. In every non-zero row, the first non-zero element is a 1
 3. The 1 in any row must be to the **left** of their counterpart in the immediate following row
- ▶ The rank of the matrix is equal to the number of non-zero rows $\rightarrow A_5$ has a rank of 2.
- ▶ Note that $r(A_5) = 2 = r(A)$ since the operations do not change the rank

Rank of a Matrix

- ▶ For a square $n \times n$ matrix, it must be that $r(A) = n$ for it to be non-singular.
- ▶ A $n \times n$ echelon matrix with no zero rows is non-singular.
- ▶ If **A is non-singular** $\Rightarrow$ **A^{-1} exists.**

Rank of a Matrix

Find the rank of each matrix.

$$A = \begin{bmatrix} 1 & 5 & 1 \\ 0 & 3 & 9 \\ -1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & -1 & -4 \\ 3 & 1 & 2 \\ 6 & 1 & 0 \end{bmatrix}$$

Rank of a Matrix

$$A = \begin{bmatrix} 1 & 5 & 1 \\ 0 & 3 & 9 \\ -1 & 0 & 0 \end{bmatrix}$$

$$R_3 = R_1 + R_3 \Rightarrow \begin{bmatrix} 1 & 5 & 1 \\ 0 & 3 & 9 \\ 0 & 5 & 1 \end{bmatrix}$$

$$R_2 = \frac{1}{3}R_2 \Rightarrow \begin{bmatrix} 1 & 5 & 1 \\ 0 & 1 & 3 \\ 0 & 5 & 1 \end{bmatrix}$$

Rank of a Matrix

$$R_3 = 5R_2 + (-1)R_3 \begin{bmatrix} 1 & 5 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix}$$

$$R_3 = \frac{1}{14}R_3 \begin{bmatrix} 1 & 5 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$r(A) = 3$$

Rank of a Matrix

$$B = \begin{bmatrix} 0 & -1 & -4 \\ 3 & 1 & 2 \\ 6 & 1 & 0 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3 \Rightarrow \begin{bmatrix} 6 & 1 & 0 \\ 3 & 1 & 2 \\ 0 & -1 & -4 \end{bmatrix}$$

$$R_1 = \frac{1}{6}R_1 \begin{bmatrix} 1 & \frac{1}{6} & 0 \\ 3 & 1 & 2 \\ 0 & -1 & -4 \end{bmatrix}$$

Rank of a Matrix

$$R_2 = 3R_1 + (-1)R_2 \Rightarrow \begin{bmatrix} 1 & \frac{1}{6} & 0 \\ 0 & \frac{-1}{2} & -2 \\ 0 & -1 & -4 \end{bmatrix}$$

$$R_2 = -2R_2 \Rightarrow \begin{bmatrix} 1 & \frac{1}{6} & 0 \\ 0 & 1 & 4 \\ 0 & -1 & -4 \end{bmatrix}$$

$$R_3 = R_2 + R_3 \Rightarrow \begin{bmatrix} 1 & \frac{1}{6} & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$r(B) = 2$$

Test of Non-Singularity by use of Determinant

Determinants

- ▶ To determine non-singularity we can make use of the concept of the *determinant*.

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- ▶ Denote the determinant of A , as $|A|$.

Determinants

- ▶ To determine non-singularity we can make use of the concept of the *determinant*.
- ▶ Denote the determinant of A , as $|A|$.
- ▶ For a 2×2 matrix, $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, its determinant is

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Determinants

For example,

$$A = \begin{bmatrix} 10 & 4 \\ 8 & 5 \end{bmatrix} \Rightarrow |A| = 5(10) - 4(8) = 50 - 32 = 18$$

or

$$B = \begin{bmatrix} 3 & 5 \\ 0 & -1 \end{bmatrix} \Rightarrow |B| = -3 + 0 = -3$$

Determinants

Consider

$$C = \begin{bmatrix} 3 & 8 \\ 3 & 8 \end{bmatrix} = \begin{bmatrix} c'_1 \\ c'_2 \end{bmatrix} \text{ and } D = \begin{bmatrix} 2 & 6 \\ 8 & 24 \end{bmatrix} = \begin{bmatrix} d'_1 \\ d'_2 \end{bmatrix}$$

- ▶ Both C and D are linearly dependent
- ▶ For $C \rightarrow c'_1 = c'_2$ and $D \rightarrow 4d'_1 = d'_2$
- ▶ Interestingly, the determinants are $|C| = 0$ and $|D| = 0$

Evaluating a Third-Order Determinant

► Suppose $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

► The determinant for A can be expressed as follows

$$|A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$= (a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32}) - (a_{12}a_{21}a_{33} - a_{12}a_{23}a_{31}) + \\ (a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31})$$

= a scalar.

Third-Order Determinant

Find the determinant of A and B .

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} -7 & 0 & 3 \\ 9 & 1 & 4 \\ 0 & 6 & 5 \end{bmatrix}$$

Third-Order Determinant

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$|A| = 2 \cdot \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 1 \cdot \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \cdot \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

$$= 2 \cdot (5 \cdot 9 - 6 \cdot 8) - 1 \cdot (4 \cdot 9 - 6 \cdot 7) + 3 \cdot (4 \cdot 8 - 5 \cdot 7)$$

$$= 2 \cdot (45 - 48) - 1 \cdot (36 - 42) + 3 \cdot (32 - 35)$$

$$= 2 \cdot (-3) - 1 \cdot (-6) + 3 \cdot (-3) = -6 + 6 - 9 = \boxed{-9}$$

Third-Order Determinant

$$B = \begin{bmatrix} -7 & 0 & 3 \\ 9 & 1 & 4 \\ 0 & 6 & 5 \end{bmatrix}$$

$$|B| = -7 \cdot \begin{vmatrix} 1 & 4 \\ 6 & 5 \end{vmatrix} - 0 \cdot \begin{vmatrix} 9 & 4 \\ 0 & 5 \end{vmatrix} + 3 \cdot \begin{vmatrix} 9 & 1 \\ 0 & 6 \end{vmatrix}$$

$$= -7 \cdot (5 - 24) + 0 + 3 \cdot (54)$$

$$= -7 \cdot (-19) + 162 = 133 + 162 = \boxed{295}$$

Sub-determinants

- ▶ To find the determinant of a 3×3 matrix we eliminated a row and column and were left with a sub-determinant of A .
- ▶ For a_{11} , we would call it the *minor* of a_{11} or $|M_{11}|$
- ▶ $\Rightarrow |M_{11}| = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$, $|M_{12}| = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$, $|M_{13}| = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$

Sub-determinants

- ▶ We can also define a cofactor

$$c_{ij} = (-1)^{i+j} |M_{ij}|$$

- ▶ Note the sign which goes in front is determined by $i + j$

Cofactors

Consider $A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$

- ▶ The minor of a_{12} is $|M_{12}| = \begin{vmatrix} 6 & 4 \\ 3 & 1 \end{vmatrix} = -6$
- ▶ Cofactor of a_{12} is $(-1)^{1+2}|M_{12}| = -(-6) = 6$
- ▶ What is the co-factor of a_{23} ?

Cofactors

Consider $A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$

- What is the co-factor of a_{23} ?

$$c_{23} = (-1)^5 \begin{vmatrix} 9 & 8 \\ 3 & 2 \end{vmatrix} = -[18 - 24] = 6$$

Higher-Order Determinants

- We can express a third-order determinants as

$$|A| = a_{11}|M_{11}| - a_{12}|M_{12}| + a_{13}|M_{13}|$$

$$= a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$$

$$= \sum_j^3 a_{1j}c_{1j} \rightarrow \text{using first row.}$$

Higher-Order Determinants

- ▶ We can express a third-order determinants as

$$|A| = a_{11}|M_{11}| - a_{12}|M_{12}| + a_{13}|M_{13}|$$

$$= a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$$

$$= \sum_j^3 a_{1j}c_{1j} \rightarrow \text{using first row.}$$

- ▶ We can expand this to a fourth order determinant

$$|B| = \sum_j^4 b_{1j}c_{1j}$$

Higher-Order Determinants

- ▶ You can do the determinant expansion using any row or column
- ▶ Try to use the easiest row or column (i.e., with most zeros or ones)

Properties of Determinants

1. The **transpose** of a matrix does not affect the value of a determinant. e.g.,

$$\begin{vmatrix} 4 & 3 \\ 5 & 6 \end{vmatrix} = \begin{vmatrix} 4 & 5 \\ 3 & 6 \end{vmatrix}$$

2. Interchanging any two rows or columns will alter the sign, but not the numerical value. e.g.,

$$\begin{vmatrix} 0 & 1 & 3 \\ 2 & 5 & 7 \\ 3 & 0 & 1 \end{vmatrix} = 0 - (2 - 21) + 3(0 - 15) = -26$$

$$C_1 \leftrightarrow C_3 \Rightarrow \begin{vmatrix} 3 & 1 & 0 \\ 7 & 5 & 2 \\ 1 & 0 & 3 \end{vmatrix} = 3(15) - (21 - 2) + 0 = +26$$

Properties of Determinants

3. Multiplication of any one row or column by a scalar k will change the value of the determinant by k -fold ($\neq$ scaling entire matrix by k).

$$\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = kad - kbc = k(ad - bc) = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

This also allows us to factor e.g.,

$$\begin{vmatrix} 15a & 7b \\ 12c & 2d \end{vmatrix} \rightarrow \text{factor out a 3 from } c_1 \Rightarrow 3 \begin{vmatrix} 5a & 7b \\ 4c & 2d \end{vmatrix}$$

$$\text{Factor out 2 from } R_2 \Rightarrow 3(2) \begin{vmatrix} 5a & 7b \\ 2c & d \end{vmatrix} = 6(5ad - 14bc)$$

Properties of Determinants

4. The addition (subtraction) of a multiple of any row to (from) another row will leave the value of the determinant unaltered. The same goes for column operations.

$$\begin{aligned} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow R_2 = R_2 + kR_1 &\implies \begin{vmatrix} a & b \\ c + ka & d + kb \end{vmatrix} \\ &= a(d + kb) - b(c + ka) = ad + akb - bc - bka \\ &= ad - bc = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \end{aligned}$$

Properties of Determinants

5. If one row (or column) is a multiple of another row (or column) $\Rightarrow$ **determinant** = 0

$$\begin{vmatrix} 2a & 2b \\ a & b \end{vmatrix} = 2ab - 2ab = 0$$

Also called a “vanishing determinant.”

Determinants and Non-Singularity

- ▶ Linear dependence $\Rightarrow$ vanishing determinant
- ▶ If $|A| \neq 0$ then
 - ▶ A is non-singular.
 - ▶ A^{-1} exists
 - ▶ $x^* = A^{-1}d \Rightarrow$ a unique solution exists.

Determinants and Non-Singularity

Does this system of equations have a unique solution?

$$7x_1 - 3x_2 - 3x_3 = 7$$

$$2x_1 + 4x_2 + x_3 = 0$$

$$-2x_2 - x_3 = 2$$

Determinants and Non-Singularity

$$7x_1 - 3x_2 - 3x_3 = 7$$

$$2x_1 + 4x_2 + x_3 = 0$$

$$-2x_2 - x_3 = 2$$

$$\begin{aligned} A &= \begin{bmatrix} 7 & -3 & -3 \\ 2 & 4 & 1 \\ 0 & -2 & -1 \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} 7 & -3 & -3 \\ 2 & 4 & 1 \\ 0 & -2 & -1 \end{vmatrix} \\ &= 7 \cdot \begin{vmatrix} 4 & 1 \\ -2 & -1 \end{vmatrix} - (-3) \cdot \begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} + (-3) \cdot \begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix} \\ &= 7 \cdot (-4 + 2) + 3 \cdot (-2) - 3 \cdot (-4) \\ &= -14 - 6 + 12 = \boxed{-8} \end{aligned}$$

Expansion of a Determinant by Alien Cofactor

- ▶ Another property: The expansion of a determinant by alien co-factors (i.e., cofactors of a “wrong” row) always yields a value of zero.

- ▶ Consider $A = \begin{bmatrix} 4 & 1 & 2 \\ 5 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$

- ▶ Using correct cofactor expansion:

$$|A| = 4[2(3) - 0] - [5(3) - 1] + 2[0 - 2] = 6$$

$$= a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$$

Expansion of a Determinant by Alien Cofactor

- ▶ What if we used *alien cofactors*?
- ▶ Suppose we used the cofactors of the **second** row

$$\begin{aligned} &= a_{11}c_{21} + a_{12}c_{22} + a_{13}c_{23} \\ &= -4 \begin{vmatrix} 1 & 2 \\ 0 & 3 \end{vmatrix} + \begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix} - 2 \begin{vmatrix} 4 & 1 \\ 1 & 0 \end{vmatrix} \\ &= -4(3) + (12 - 2) - 2(-1) = -12 + 10 + 2 = 0 \end{aligned}$$

Expansion of a Determinant by Alien Cofactor

► More generally, $\sum_{j=1}^3 a_{1j}c_{2j}$

$$\begin{aligned} &= -a_{11} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{12} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{13} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ &= -a_{11}[a_{12}a_{33} - a_{13}a_{32}] + a_{12}[a_{11}a_{33} - a_{13}a_{31}] \\ &\quad - a_{13}[a_{11}a_{32} - a_{12}a_{31}] \\ &= 0 \end{aligned}$$

Expansion of a Determinant by Alien Cofactor

► More generally, $\sum_{j=1}^3 a_{1j}c_{2j}$

$$\begin{aligned} &= -a_{11} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{12} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{13} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ &= -a_{11}[a_{12}a_{33} - a_{13}a_{32}] + a_{12}[a_{11}a_{33} - a_{13}a_{31}] \\ &\quad - a_{13}[a_{11}a_{32} - a_{12}a_{31}] \\ &= 0 \end{aligned}$$

► This applies to any alien cofactor. Generally,

$$\sum_{j=1}^n a_{ij}c_{i'j} = 0 \text{ for } i \neq i'$$

Finding the Inverse of a Matrix

Finding the Inverse

$$\text{Assume } A_{n \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

- ▶ Since each element of A has a cofactor, it is possible to form a matrix of cofactors.

$$C = [c_{ij}]$$

- ▶ However, we will need the transpose of C , C' , called the *adjoint* of A .

Finding the Inverse

$$A_{n \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

$$C'_{n \times n} = Adj(A) = \begin{bmatrix} c_{11} & c_{21} & \cdots & c_{n1} \\ c_{12} & c_{22} & \cdots & c_{n2} \\ \cdots & \cdots & \cdots & \cdots \\ c_{1n} & c_{2n} & \cdots & c_{nn} \end{bmatrix}$$

- ▶ Note that A and C' are conformable for multiplication.
- ▶ Their product AC' is another $n \times n$ matrix in which each element is the sum of products.

Finding the Inverse

$$AC'_{n \times n} = \begin{bmatrix} \sum_{j=1}^n a_{1j}c_{1j} & \sum_{j=1}^n a_{1j}c_{2j} & \cdots & \sum_{j=1}^n a_{1j}c_{nj} \\ \sum_{j=1}^n a_{2j}c_{1j} & \sum_{j=1}^n a_{2j}c_{2j} & \cdots & \sum_{j=1}^n a_{2j}c_{nj} \\ \cdots & \cdots & \cdots & \cdots \\ \sum_{j=1}^n a_{nj}c_{1j} & \sum_{j=1}^n a_{nj}c_{2j} & \cdots & \sum_{j=1}^n a_{nj}c_{nj} \end{bmatrix}$$

- ▶ Interestingly, the off diagonals are using alien cofactor expansion.
- ▶ The diagonals are using the same row cofactors, which provide us with the determinant of A

Finding the Inverse

- ▶ Since the off diagonals are using alien cofactor expansion these are all zeros.

$$AC'_{n \times n} = \begin{bmatrix} \sum_{j=1}^n a_{1j}c_{1j} & 0 & \cdots & 0 \\ 0 & \sum_{j=1}^n a_{2j}c_{2j} & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \sum_{j=1}^n a_{nj}c_{nj} \end{bmatrix}$$

Finding the Inverse

- ▶ The diagonals are the determinants of A

$$AC'_{n \times n} = \begin{bmatrix} |A| & 0 & \cdots & 0 \\ 0 & |A| & \cdots & 0 \\ \vdots & \cdots & \cdots & \vdots \\ 0 & 0 & \cdots & |A| \end{bmatrix}$$

- ▶ Since $|A|$ is a non-zero scalar, we can factor

$$AC'_{n \times n} = |A| \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \cdots & \cdots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

- ▶ Notice that we are left with the determinant of A times an identity matrix.

Finding the Inverse

$$AC' = |A| \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = |A| \cdot I_n$$
$$\Rightarrow AC' = |A| \cdot I_n$$

$$\Rightarrow \frac{AC'}{|A|} = I_n \Rightarrow A \cdot \frac{C'}{|A|} = I_n$$

Now pre-multiply by A^{-1} :

$$A^{-1}A \cdot \frac{C'}{|A|} = A^{-1}I_n$$

Finding the Inverse

$$\Rightarrow \frac{C'}{|A|} = A^{-1}$$

- ▶ $\frac{C'}{|A|}$ is equal to the inverse of A .

Steps to find Inverse

1. Confirm $|A| \neq 0$
2. Find cofactors of all elements of A and arrange as a matrix $C = [c_{ij}]$
3. Take the transpose of C to get $Adj(A) = C'$
4. Divide the $Adj(A)$ by the determinant of A , $|A|$.

Finding the Inverse

Suppose we have $A = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$

1. $|A| = 3(0) - 2(1) = -2 \neq 0$

Finding the Inverse

Suppose we have $A = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$

1. $|A| = 3(0) - 2(1) = -2 \neq 0$
2. Construct matrix of cofactors (don't forget about signs)

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & -1 \\ -2 & 3 \end{bmatrix}$$

Finding the Inverse

Suppose we have $A = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$

1. $|A| = 3(0) - 2(1) = -2 \neq 0$
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$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & -1 \\ -2 & 3 \end{bmatrix}$$

3. Take transpose of C

$$C' = \begin{bmatrix} 0 & -2 \\ -1 & 3 \end{bmatrix} = \text{Adj}(A)$$

Finding the Inverse

Suppose we have $A = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$

1. $|A| = 3(0) - 2(1) = -2 \neq 0$
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$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & -1 \\ -2 & 3 \end{bmatrix}$$

3. Take transpose of C

$$C' = \begin{bmatrix} 0 & -2 \\ -1 & 3 \end{bmatrix} = \text{Adj}(A)$$

4. $A^{-1} = \frac{\text{Adj}(A)}{|A|} = \frac{-1}{2} \begin{bmatrix} 0 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{-3}{2} \end{bmatrix}$

Finding the Inverse

1. Find the inverse of $B = \begin{bmatrix} 4 & 1 & -1 \\ 0 & 3 & 2 \\ 3 & 0 & 7 \end{bmatrix}$

2. Solve the system of equations using matrix operations

$$4x + 3y = 28$$

$$2x + 5y = 42$$

3. Solve the system of equations using matrix operations

$$4x_1 + x_2 - 5x_3 = 8$$

$$-2x_1 + 3x_2 + x_3 = 12$$

$$3x_1 - x_2 + 4x_3 = 5$$

Cramer's Rule

Cramer's Rule

We have that $x^* = A^{-1}d = \frac{1}{|A|} \cdot Adj(A) \cdot d$

$$\Rightarrow \begin{bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_n^* \end{bmatrix} = \frac{1}{|A|} \begin{bmatrix} c_{11} & c_{21} & \cdots & c_{n1} \\ c_{12} & c_{22} & \cdots & c_{n2} \\ \cdots & \cdots & \cdots & \cdots \\ c_{1n} & c_{2n} & \cdots & c_{nn} \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{bmatrix}$$

$n \times n$ $n \times 1$

► Resulting matrix will be $n \times 1$

Cramer's Rule

$$\Rightarrow \begin{bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_n^* \end{bmatrix} = \frac{1}{|A|} \begin{bmatrix} d_1 c_{11} + d_2 c_{21} + \cdots + d_n c_{n1} \\ d_1 c_{12} + d_2 c_{22} + \cdots + d_n c_{n2} \\ \vdots \\ d_1 c_{1n} + d_2 c_{2n} + \cdots + d_n c_{nn} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_n^* \end{bmatrix} = \frac{1}{|A|} \begin{bmatrix} \sum_{i=1}^n d_i c_{i1} \\ \sum_{i=1}^n d_i c_{i2} \\ \vdots \\ \sum_{i=1}^n d_i c_{in} \end{bmatrix}$$

Cramer's Rule

$$\begin{bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_n^* \end{bmatrix} = \frac{1}{|A|} \begin{bmatrix} \sum_{i=1}^n d_i c_{i1} \\ \sum_{i=1}^n d_i c_{i2} \\ \vdots \\ \sum_{i=1}^n d_i c_{in} \end{bmatrix}$$

- If we equate the corresponding elements

$$x_1^* = \frac{1}{|A|} \sum_{i=1}^n d_i c_{i1}$$

$$x_2^* = \frac{1}{|A|} \sum_{i=1}^n d_i c_{i2}$$

$$\vdots$$

$$x_n^* = \frac{1}{|A|} \sum_{i=1}^n d_i c_{in}$$

Cramer's Rule

$$x_n^* = \frac{1}{|A|} \sum_{i=1}^n d_i c_{in}$$

- ▶ These terms are similar to the following expansion (as if we were finding $|A|$ using the first column)

$$\sum_{i=1}^n a_{i1} c_{i1}$$

Cramer's Rule

$$x_n^* = \frac{1}{|A|} \sum_{i=1}^n d_i c_{in}$$

- ▶ These terms are similar to the following expansion (as if we were finding $|A|$ using the first column)

$$\sum_{i=1}^n a_{i1} c_{i1}$$

- ▶ The difference is that the first column is being replaced with vector d , but other columns are kept intact.

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$$\Rightarrow x_1^* = \frac{|A_1|}{|A|}$$

- ▶ Similarly, if we replace the second column of A by the column vector d (and retain all other columns)

$$\Rightarrow x_2^* = \frac{|A_2|}{|A|}$$

Cramer's Rule

We can generalize this procedure. To find the solution to the j^{th} variable x_j^*

1. Replace the j^{th} column of the determinant $|A|$ by the constant terms $d_1, d_2, \dots, d_n$
2. Get new determinant $|A_j|$
3. Divide $|A_j|$ by $|A|$

$$\Rightarrow x_j^* = \frac{|A_j|}{|A|} = \frac{1}{|A|} \begin{vmatrix} a_{11} & a_{12} & \cdots & d_1 & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & d_2 & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & d_n & \cdots & a_{nn} \end{vmatrix}$$

- This is Cramer's rule. Note that matrix inversion gives us solutions to all x^* s, whereas with Cramer's rule you have to go one by one.

Cramer's Rule

Use Cramer's rule to solve the following systems of equations

1.
$$\begin{aligned} 5x_1 + 3x_2 &= 30 \\ 6x_1 - 2x_2 &= 8 \end{aligned}$$

2.
$$\begin{aligned} 7x_1 - x_2 - x_3 &= 0 \\ 10x_1 - 2x_2 + x_3 &= 8 \\ 6x_1 + 3x_2 - 2x_3 &= 7 \end{aligned}$$

Finding Inverses by Row Operations

Inverse by Row Operations

Recall our elementary row operations

1. Interchanging any two rows
2. Multiply each element of a row by any scalar $k \neq 0$
3. Addition of k times any row to another row.

Steps to Invert $n \times n$ Matrix

1. Form $n \times 2n$ matrix $A : I$
 - Write down n columns of A followed by n columns of I
2. Apply elementary row operations until you end up with $n \times 2n$ matrix $I : B$ whose **first n columns** are the columns of the identity matrix I
3. $\Rightarrow B = A^{-1}$
4. If it is impossible to perform such operations $\Rightarrow A$ has no inverse.

Inverse by Row Operations

Find the inverse of $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 3 \end{bmatrix}$

1. Will there be an inverse?

$$|A| = (9 - 16) - 3(3 - 4) + 3(4 - 3) = -1 \checkmark$$

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2.

$$[A : I] = \begin{bmatrix} 1 & 3 & 3 & : & 1 & 0 & 0 \\ 1 & 3 & 4 & : & 0 & 1 & 0 \\ 1 & 4 & 3 & : & 0 & 0 & 1 \end{bmatrix}$$

Inverse by Row Operations

$$3. R_3 = R_1 - R_3 \Rightarrow \begin{bmatrix} 1 & 3 & 3 & \vdots & 1 & 0 & 0 \\ 1 & 3 & 4 & \vdots & 0 & 1 & 0 \\ 0 & -1 & 0 & \vdots & 1 & 0 & -1 \end{bmatrix}$$

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$$R_2 = R_1 - R_2 \Rightarrow \begin{bmatrix} 1 & 3 & 3 & \vdots & 1 & 0 & 0 \\ 0 & 0 & -1 & \vdots & 1 & -1 & 0 \\ 0 & -1 & 0 & \vdots & 1 & 0 & -1 \end{bmatrix}$$

Inverse by Row Operations

$$R_1 = 3R_2 + R_1 \Rightarrow \begin{bmatrix} 1 & 3 & 0 & \vdots & 4 & -3 & 0 \\ 0 & 0 & -1 & \vdots & 1 & -1 & 0 \\ 0 & -1 & 0 & \vdots & 1 & 0 & -1 \end{bmatrix}$$

$$R_2 \circlearrowleft R_3 \Rightarrow \begin{bmatrix} 1 & 3 & 0 & \vdots & 4 & -3 & 0 \\ 0 & -1 & 0 & \vdots & 1 & 0 & -1 \\ 0 & 0 & -1 & \vdots & 1 & -1 & 0 \end{bmatrix}$$

$$\begin{matrix} R_2 = -1 \cdot R_2 \\ R_3 = -1 \cdot R_3 \end{matrix} \Rightarrow \begin{bmatrix} 1 & 3 & 0 & \vdots & 4 & -3 & 0 \\ 0 & 1 & 0 & \vdots & -1 & 0 & 1 \\ 0 & 0 & 1 & \vdots & -1 & 1 & 0 \end{bmatrix}$$

Inverse by Row Operations

$$R_1 = -3R_2 + R_1 \Rightarrow \begin{bmatrix} 1 & 0 & 0 & \vdots & 7 & -3 & -3 \\ 0 & 1 & 0 & \vdots & -1 & 0 & 1 \\ 0 & 0 & 1 & \vdots & -1 & 1 & 0 \end{bmatrix}$$
$$\Rightarrow A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

- If we had column d , we could find $x^* = A^{-1}d$

Inverse by Row Operations

1. Find inverse using row operations of $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
2. Find inverse using row operations of $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$
3. Solve this system of equations using row operations to find inverse

$$2x + 4y - 2z = 6$$

$$4x + 5y - 3z = 13$$

$$6x + 7y - 5z = 19$$

Homogeneous Equations Systems

- ▶ Consider the system $Ax = 0$ (0 is a vector) called a Homogeneous system of equations.
- ▶ If A is non-singular, a homogeneous equation system can only yield a trivial solution $x_1^* = x_2^* = x_3^* = \dots x_n^* = 0$
- ▶ This follows from

$$x^* = A^{-1}d \Rightarrow x^* = A^{-1}0 = 0$$

- ▶ Or we can derive this from Cramer's rule

$$x_j^* = \frac{|A_j|}{|A|} = \frac{0}{|A|} = 0$$